॥ वसुधैव कुटुम्बकम् ॥

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)
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ASSIGNMENT No: 1

TITLE: Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Problem Statement: Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

AIM: To install Java JDK 21, write a Java program using a static method to calculate the square of a number, compile it, and execute it using the command line.

Exercise:

1. **Explain the main concept used in Java environment setup, variables and methods.**

   Answer:

   Java environment setup involves installing the Java Development Kit (JDK), setting the required environment variables (such as PATH), writing the program in a .java file, compiling it using the javac command, and running it using the java command.

   Variables are used to store data such as numbers, text, and other values.

   Methods are blocks of code that perform specific tasks. They improve code reusability, reduce duplication, and make programs easier to understand and maintain.

2. What are the advantages of using Java variables and methods?

Answer:

- Variables help store and manage data efficiently.
- Methods reduce code duplication by promoting code reuse.
- They improve the readability and maintainability of programs.
- Methods make the program modular and easier to debug.
- Variables make data handling simple and organized.

3. How can this implementation be improved?

Answer:

- The program can be improved by:
- Accepting user input instead of using fixed values.
- Adding input validation and exception handling.
- Using meaningful variable names and proper comments.
- Following Java coding standards for better readability.
- Dividing the program into smaller methods for better organization.

4. What are the real-world applications?

Answer:

Java is widely used in:

- Banking and financial applications
- Android mobile application development
- Web application development
- Desktop software development
- Enterprise applications
- Cloud-based applications
- E-commerce platforms
- Big Data and distributed systems
- Scientific and educational software

Source code:

Sumit.java

File Edit View

```java
public class Main {

    // Method
    static int square(int number) {
        return number * number;
    }

    public static void main(String[] args) {

        // Variables
        String name = "Sumit";
        int age = 20;
        double salary = 55000.50;
        int result = square(6);

        System.out.println("Name     : " + name);
        System.out.println("Age      : " + age);
        System.out.println("Salary   : " + salary);
        System.out.println("Square of 6 : " + result);
    }
}
```

Output:

```
C:\Users\CL408_14\Desktop\sau>java Main
Name     : Sumit
Age      : 20
Salary   : 55000.5
Square of 6 : 36

C:\Users\CL408_14\Desktop\sau>
```

Conclusion:

Through this program, I learned how to create and use methods in Java, declare different types of variables, and perform simple arithmetic operations by passing values to methods. I also understood how to compile and execute a Java program using the command line (javac and java). This experiment showed how methods help make code more organized, reusable, and easier to maintain.